

ZIAD AHMED ABDELHADY

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PROFESSIONAL SUMMARY

Results-driven Software Engineer with expertise in full-stack development, system integration, and AI-powered automation. Specialized in building production-grade web portals, Odoo ERP customization, and enterprise system integrations (AEOS access control). Proven track record in delivering business-critical solutions that reduce operational time by up to 90% through intelligent automation. Certified Professional Scrum Product Owner (PSPO I) with strong stakeholder management and cost-benefit analysis capabilities. Experienced in bridging technical execution with business value through data-driven automation consulting.

PROFESSIONAL EXPERIENCE

Hassan Allam - KODE Sports Club | Cairo, Egypt

Software Engineer & AI Automation Analyst | September 2025 – Present

- Architected and deployed multiple production-grade web portals and systems using Node.js, Next.js, TypeScript, and React, solving critical business challenges in asset tracking, security operations, and member access management
- KODE Asset Management System: Full-stack application for employee onboarding and device/license tracking, reducing annual inventory check time by 75%. Used by Technology and HR departments (Tech: React, Tailwind CSS, Node.js, Nest.js, PostgreSQL, Prisma ORM)
- Face ID Portal: Integrated web portal with Nedap AEOS backend for automated face image uploads to club's face recognition system using base64 conversion, streamlining member biometric enrollment and access control
- KODE Security Portal: Real-time member monitoring system integrated with AEOS backend via SOAP API service. Displays member entry images, tracks live member counts across club locations, and validates family member access for enhanced gate security and access integrity
- Task Management System: Implemented Scrum-based backlog system for department workflow optimization, improving team collaboration and project delivery timelines
- Developed Python automation scripts for data cleaning and processing, reducing data preparation time by up to 90% and enabling faster business intelligence reporting
- Built REST API integrations and data analysis pipelines connecting multiple enterprise systems (AEOS, Odoo, internal portals) for seamless data flow and operational efficiency
- Odoo ERP Development: Customized and extended Odoo 17 modules for business-specific workflows, including custom reports, automated processes, and system integrations
- AI Automation Analyst: Conducted stakeholder meetings with department managers to identify automation opportunities, prepared detailed proof-of-concepts (POCs) and professional reports analyzing cost savings and time management improvements for potential business solutions

- Managed complete product lifecycle from development to production deployment, ensuring seamless transitions and system reliability. Applied PSPO I principles to prioritize features based on ROI and minimize technical debt
- Delivered professional business reports, technical documentation, and stakeholder presentations demonstrating solution impact and business value

Customer Experience Associate | July 2025 – September 2025

- Designed comprehensive training program for new CX agents, creating structured onboarding materials that reduced ramp-up time and improved team performance metrics
- Authored official communication documentation and guidelines for the CX office, establishing standardized processes that enhanced team collaboration and service consistency

TECHNICAL SKILLS

- **Programming Languages:** JavaScript, TypeScript, Python, HTML, CSS, SQL
- **Frontend Development:** React.js, Next.js, Tailwind CSS
- **Backend Development:** Node.js, Nest.js, Express.js, REST APIs, SOAP APIs
- **Databases:** PostgreSQL, MongoDB, Prisma ORM
- **ERP Systems:** Odoo 17 (Custom Module Development, Integration, Automation)
- **System Integration:** Nedap AEOS (SOAP API), Base64 encoding, Access Control Systems
- **Data Engineering:** Python data processing, automation scripts, ETL pipelines, data cleaning
- **AI/ML:** MATLAB, Artificial Neural Networks (ANN), Machine Learning algorithms
- **Product Management:** Professional Scrum Product Owner (PSPO I), Agile/Scrum methodologies, Stakeholder management, ROI analysis
- **Development Tools:** Git, GitHub, VS Code, NPM, MATLAB

KEY PROJECTS

KODE Asset Management System

Full-stack employee onboarding and asset tracking system deployed on company's internal network. Manages device assignments, software licenses, and equipment inventory for Technology and HR departments. Reduced annual inventory check time by 75%.

Tech Stack: React, Tailwind CSS, Node.js, Nest.js, PostgreSQL, Prisma ORM

Face ID Portal - AEOS Integration

Web-based portal integrated with Nedap AEOS access control backend for automated face image uploads. Converts and uploads member photos using base64 encoding for face recognition enrollment, streamlining biometric access management.

Tech Stack: React, Node.js, AEOS API, Base64 encoding

KODE Security Portal - AEOS SOAP Integration

Real-time security monitoring system integrated with AEOS backend via SOAP API. Displays member entry images at access points, tracks live member counts across multiple club locations based on access logs, and validates family member relationships to ensure access integrity at member gates.

Tech Stack: React, Node.js, SOAP API, AEOS backend, Real-time data processing

Galassia E-Commerce Platform

Luxury e-commerce platform for gemstones and fine jewelry. Features comprehensive admin panel for product/category management and analytics dashboard tracking users, orders, and revenue metrics.

Tech Stack: React, Tailwind CSS, Node.js, Express.js, MongoDB

Demo URL: <https://galassia-theta.vercel.app/>

EDUCATION

Bachelor of Civil Engineering with Honors

UNISEL University | Selangor, Malaysia | GPA: 3.2 | September 2024

Thesis: "Structural Damage Detection of Plain Stress using Inverse Frequency Analysis incorporating Artificial Neural Network (ANN)" | Grade: A+

- Developed AI-powered structural damage detection system using Artificial Neural Networks in MATLAB over one-year research period
- Applied machine learning algorithms to analyze complex stress patterns and predict structural failures with high accuracy
- Demonstrated advanced problem-solving and technical proficiency in AI/ML, catalyzing transition to software development career

Certified Full Stack Developer (MERN Stack)

Route Academy | Cairo, Egypt | February 2025 – January 2026

- Frontend: TypeScript, React.js, Next.js
- Backend: Node.js, Express.js, Nest.js
- Databases: PostgreSQL, MongoDB